Micro-Macro Dynamics Between Changes in Consumer Behavior and Market-Wide Shortages During a Crisis

An Agent-Based Simulation Informed by Survey Data

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1 Abstract

To be drafted

2 Introduction

During acute crises such as pandemics, natural disasters, or supply warnings, consumers frequently rush to buy daily necessities, emptying retail shelves across a region or country within days (Keane & Neal, 2021; Yuen et al., 2020). Many of these sudden shortages occur even when aggregate production and supply chains remain sufficient, or before any physical disruption reaches retail stores (Bentall et al., 2021). In Japan, for example, toilet paper disappeared from stores nationwide in early 2020 despite abundant domestic production and inventory (Iizuka et al., 2022). Because such demand-driven runs disrupt retail distribution and leave many households unable to obtain essentials, a large body of research, especially since the COVID-19 pandemic, has sought to explain why panic buying occurs and how these shortages unfold.

Most empirical studies approach panic buying at the individual level, using cross-sectional surveys to test which psychological states, perceptions, and traits predict self-reported purchasing or stockpiling (Yuen et al., 2020). Across this literature, higher buying intentions are associated with negative emotions and threat appraisals such as anxiety, cyberchondria, and anticipated regret (Bentall et al., 2021; Chua et al., 2021; Laato et al., 2020), perceived product scarcity and social or media cues (Chua et al., 2021; Prentice et al., 2022), and personality traits such as conscientiousness, emotionality, and paranoia (Bentall et al., 2021; Garbe et al., 2020). However, explaining a market-wide shortage solely through individual psychological predictors has three limitations. First, stable dispositions and general crisis distress explain which individuals report slightly higher buying tendencies at a given moment, but they cannot explain why a run on a specific commodity erupts and ends within days. Indeed, even after severe national shortages in the United Kingdom and Ireland, nationally representative surveys showed that roughly three-quarters of adults reported minimal over-purchasing (Bentall et al., 2021), and extra buying during crises often reflects ordinary precaution rather than acute emotional panic (Lindell, 2024; Ntontis et al., 2022). Second, survey models (often using a stimulus-organism-response framework; Chua et al., 2021; Laato et al., 2020) typically treat scarcity cues, empty shelves, or peer buying as an external stimulus that predicts an individual's own purchase intention. While this captures how a single consumer reacts to an existing cue, it takes the scarcity cue as given rather than explaining how consumers' own actions create it. Third, by ending at individual purchasing, these studies treat consumers only as passive receivers of crisis and shortage information, overlooking that consumers also act as transmitters who share shortage information with others.

A second stream of empirical research examines panic buying at the market level using retail transaction records or social media logs. High-frequency household scanner studies of the 2011 Tohoku earthquake in Japan (Hori & Iwamoto, 2014) and the early 2020 COVID-19 wave in the United Kingdom (O'Connell et al., 2021) show that aggregate demand spikes were driven mainly by an increase in the share of households choosing to buy, with only small increases in the quantity purchased per trip and few extreme hoarders. However, transaction logs record only the

aggregate buying outcome; they cannot observe the interpersonal interactions and information sharing that prompt many ordinary households to shop within the same short window. Studies that combine social media data with retail sales attempt to capture this informational process, but they observe only macro trends and fit untested micro explanations to them. For instance, Iizuka et al. (2022) regressed daily supermarket toilet-paper sales during Japan's 2020 shortage directly on daily estimated Twitter views, concluding from their co-movement that widely shared corrective posts drove excessive purchasing through pluralistic ignorance. Yet mapping tweet volume directly onto sales treats daily purchases as a simple function of online views while ignoring physical store stockouts and household saturation, which naturally cause sales to drop after a spike once shelves are empty and early buyers are satisfied. It also infers individual psychology from aggregate time series without testing how consumers actually react to or share shortage information, especially through private conversations outside public social media.

A third stream of research uses theoretical or simulation models to examine how individual actions aggregate into market-wide shortages. For example, economic equilibrium models show that even fully rational consumers who anticipate a temporary rise in shopping costs can trigger hoarding spirals and severe shortages through congestion externalities (Noda & Teramoto, 2024). Other studies use agent-based or network simulations to model how panic buying spreads, either by adapting epidemic contagion rules across social networks (Fu et al., 2021) or by predicting individual purchase probabilities from survey regressions (Dulam et al., 2021). However, existing models typically rely on stylized utility calculations or unverified contagion assumptions, and even models informed by survey data focus on purchasing alone without empirically measuring how consumers share shortage information with others.

[NOTE: The paragraph above, on theoretical and simulation models, is temporary. Around 80 theoretical and simulation papers were identified in the literature search and are still under review, so the summary of modeling approaches and the specific gap stated here may be updated after checking those papers in full.]

Addressing these limitations requires examining the micro-macro process through which individual actions generate a collective outcome. From a micro-macro perspective in social psychology, a market-wide shortage is an emergent, unintended consequence of interdependent actions rather than a simple aggregation of individual traits or panic motives (Schelling, 1978; Yamagishi, 1992). Because consumers do not act in isolation, one person's observable behavior alters the environment faced by others in two ways (Yamagishi, 1992): purchasing physically reduces local store stock, creating an objective empty-shelf cue for subsequent shoppers, while sharing information transmits shortage cues to others before they visit a store, shaping their social reality. When purchasing and information sharing feed back into each other across consumers over time, even small shifts in individual tendencies, which look modest in a cross-sectional survey, can accumulate nonlinearly into a sudden market-wide shortage. Explaining how a demand-driven shortage occurs therefore requires combining empirical data on how individuals both purchase and share information in response to shortage cues with a dynamic model of how those interdependent behaviors accumulate at the market level.

[NOTE: The paragraph above, on the micro-macro perspective, outlines the general methodological argument I plan to make for bridging the micro-macro gap, currently anchored in Yam-

agishi (1992) and Schelling (1978). I am not yet sure whether this wording and framing are good enough or which standard methodological references are best to cite here, so I would appreciate your advice on this part.]

Guided by this perspective, the present thesis combines empirical surveys with an agent-based simulation to examine how individual responses to shortage information can produce a market-wide shortage under minimal, empirically grounded assumptions. Rather than assuming how consumers act in a simulation from the outset, we first used surveys to investigate how consumers respond when they encounter shortage information, either directly at a nearly empty store shelf or indirectly through a social media post, measuring both their purchase intention and their information-sharing intention and examining how these intentions relate to their subjective prospects about future supply. Study 1 was an exploratory survey conducted after the 2024–2025 rice shortage in Japan to examine the structure of consumers' purchasing and information-sharing intentions and their associations with manipulated and measured subjective prospects for rice supply. Study 2 was a preregistered conceptual replication with two other daily necessities, toilet paper and laundry detergent, conducted during the 2026 oil-supply crisis to test whether the behavioral structure and correlations observed in Study 1 generalize across commodities and crises. Finally, building on the behavioral structure and the modest individual associations found in Studies 1 and 2, the simulation section constructs an agent-based model to test whether small changes in individual purchasing and information-sharing behavior can interact across consumers to generate rapid, market-wide depletion.

3 Study 1: Rice Supply News and Responses to Shortage Information

Study 1 examined how consumers respond to shortage information of a staple food. Rice is a staple food in Japanese households, and Japanese consumers experienced a market-wide rice shortage from 2024 to 2025 (i.e., the Reiwa rice crisis). Japan's Ministry of Agriculture, Forestry and Fisheries (MAFF) acknowledged a rice shortage in early summer 2024, which was driven by increasing demand from inbound tourism and a supply shortage caused by extreme weather in 2023 (Ohinata, 2026). Against this backdrop, consumers sharply increased their rice purchases in response to the special advisory the Japan Meteorological Agency announced on August 8, 2024, warning of an increased risk of a major earthquake (the advisory officially known as the "Nankai Trough Earthquake Extra Information"). This led to severe disruptions in rice distribution across Japan in August and September 2024 (Fukutomi, 2024). This study was conducted in January 2026, while people's memory was relatively fresh.

We expected that exposure to newspaper articles reporting positive or negative prospects for future rice supply would affect consumers' purchase intention. To test this, we manipulated prospects for future rice supply by presenting real newspaper articles (see Appendix) reporting either negative or positive analyses about the future supply. We then measured purchase and information-sharing intentions in two contexts: participants learned of a shortage directly (e.g., during shopping) or indirectly (e.g., from a social media post). The study was exploratory and was not preregistered.

3.1 Method

We recruited Japanese adults through a Japanese survey company, Cross Marketing Inc. (`https://www.cross-m.co.jp/en`), in January 2026 and analyzed 284 of 575 responses (46.8% men; age $M = 45.28$, $SD = 8.87$; exclusions in Appendix). The study was approved by the Ethics Committee of the Graduate School of Humanities and Sociology, University of Tokyo (UTSP-25047).

Participants were randomly assigned to read two real newspaper articles that were either negative ("Shinmai, Zōsanka no Neagari," 2025; Yasuda, 2025) or positive ("Kome Kakaku ni Sakiyasukan," 2025; Nakazawa, 2025) about the future supply of rice ($n = 165$ and 119, respectively). We presented excerpts rather than full articles, selecting sentences about the future supply of rice without changing their wording, so that the two articles in each condition totaled about 540 Japanese characters (542 negative, 536 positive).

They then rated their subjective prospects for future rice supply and reported their purchase intention and information-sharing intention in two contexts (see Appendix): finding the rice shelf of their usual supermarket almost empty (*direct observation*) and seeing a social media post with a photo of such a shelf (*indirect observation*). The direct/indirect observation context was a within-participant condition (the order of these two contexts was counterbalanced). These two contexts were manipulated by an AI-generated picture corresponding to each context (participants were explicitly told that the pictures were generated using AI; Figure 1).



Figure 1: Pictures shown in the direct observation context (left) and the indirect observation context (right) of Study 1. Both pictures were generated using AI, and participants were told so.

3.1.1 Measures

Subjective Prospects. Participants' subjective prospects for future rice supply were measured with three 5-point scale items: how do you think the supply will change (1 = *very unstable* to 5 = *very stable*), how do you think about it (1 = *very pessimistic* to 5 = *very optimistic*), and how likely do you think a shortage like the Reiwa rice crisis will happen again (1 = *very low* to 5 = *very high*). The first and second item scores were reverse-coded so that higher values indicate more negative prospects, and we aggregated the three item scores confirming their sufficient

internal consistency (Cronbach's $\alpha = .82$).

Purchase Intention. Participants' purchase intention in response to shortage information was measured in each observation context with three 5-point scale items (1 = *definitely not* to 5 = *definitely*; Table 5). In the direct observation context, participants rated whether they would buy the remaining rice at the store when they had rice left at home for a few days, for more than a week, or for more than a month. In the indirect observation context, they rated whether they would go to buy rice right away under the same three levels of household stock.

Information-sharing Intention. Participants' information-sharing intention was measured in each observation context with three 5-point scale items on the same scale (1 = *definitely not* to 5 = *definitely*; Table 5). Two items measured private information-sharing intention: telling housemates, best friends or romantic partner about the shortage information (close relationship item) and telling acquaintances or friends about it (casual relationship item). These two types of private information sharing intention were used in both direct and indirect observation contexts.

One item measured public information-sharing intention: posting about the store shortage to many unspecified people on social media in the direct observation context. In the indirect observation context, this item asked re-posting intention.

3.2 Results

3.2.1 Behavioral variables

Because the behavioral items (i.e., items to measure purchase intentions and information-sharing intentions) covered various types of responses to shortage information, we first conducted an exploratory factor analysis (minimum residual extraction with varimax rotation). Three eigenvalues exceeded 1 (4.96, 2.26 and 1.37; the fourth was 0.92), and the three factors explained 62.7% of the total variance (Table 1; see Appendix A.2.3 for robustness checks).

The first factor was related to purchase intention. Regardless of whether participants obtained the shortage information directly or indirectly, and regardless of how much rice they had at home, six items related to purchase intention highly loaded on the first factor. The second factor was related to private information-sharing intention. Four items about privately sharing the shortage information (i.e., sharing with either close or casual relationship partners in either direct or indirect observation context) highly loaded on the second factor. The third factor was related to public information-sharing intention. Two items about publicly sharing the shortage information on social media (i.e., either in direct or indirect observation context) highly loaded on the third factor.

Following the observed factor structure, we aggregated the six purchase intention item scores (Cronbach's $\alpha = .87$), four private information-sharing item scores ($\alpha = .86$), and two public information-sharing item scores ($\alpha = .88$; $r = .79$ between the two items), respectively.

3.2.2 Effect of the news condition

The manipulation was successful. Participants in the negative prospects condition ($M = 4.05$, $SD = 0.66$) held more negative subjective prospects about the future supply of rice than participants

Table 1: Varimax-rotated loadings of the twelve behavioral items in Study 1 (rice, $N = 284$). Minimum residual extraction; loadings $\geq .40$ in bold. Full item wording is given in Table 5.

Item	Context	Purchase	Private	Public
Buy: a few days left	Direct	**.54**	.29	−.27
Buy: a week left	Direct	**.79**	.23	−.20
Buy: a month left	Direct	**.74**	.10	.14
Go to buy: a few days	Indirect	**.63**	.35	−.05
Go to buy: a week	Indirect	**.80**	.23	.04
Go to buy: a month	Indirect	**.72**	.10	.28
Private sharing: close	Direct	.22	**.79**	−.04
Private sharing: casual	Direct	.12	**.75**	.26
Private sharing: close	Indirect	.35	**.68**	.12
Private sharing: casual	Indirect	.26	**.66**	.36
Public sharing	Direct	−.01	.17	**.80**
Public sharing	Indirect	.02	.15	**.85**
% variance		27.1	20.7	14.9

in the positive prospects condition ($M = 3.12$, $SD = 0.76$): $t(231) = 10.75$, $p < .001$, $d = 1.32$.

However, the prospects manipulation did not have significant effects on purchase intention (M_{neg} [SD] = 2.79 [0.83] vs. M_{pos} [SD] = 2.73 [0.78], t[263] = 0.64, $p = .521$, $d = 0.08$) and public information-sharing intention (M_{neg} [SD] = 1.72 [0.85] vs. M_{pos} [SD] = 1.70 [0.75], t[270] = 0.18, $p = .853$, $d = 0.02$; see Table 8 in the Appendix). The effect was significant for private information-sharing intention (M_{neg} [SD] = 2.91 [0.88] vs. M_{pos} [SD] = 2.70 [0.85], t[258] = 2.02, $p = .044$, $d = 0.24$).

3.2.3 Subjective prospects and behavioral intentions

For exploratory purposes, we computed the Pearson's product-moment correlations between subjective prospects and behavioral intention scores collapsing the newspaper conditions (Table 9 in the Appendix gives the full correlation matrix). The subjective prospects × purchase intention correlation was significant (r[282] = .18, 95% CI [.07, .29], $p = .002$), suggesting that participants with more negative subjective prospects were more willing to buy rice. The subjective prospects × private information-sharing intention correlation was also significant (r[282] = .18, 95% CI [.06, .29], $p = .003$), suggesting that participants with more negative subjective prospects were more willing to share the shortage information privately. The subjective prospects × public information-sharing intention correlation was not significant (r[282] = .05, 95% CI [−.07, .16], $p = .420$). The correlations did not differ between the news conditions (Fisher's r-to-z test; Table 10 in the Appendix).

3.3 Discussion

As intended, participants who read negative articles reported more negative prospects about the supply of rice, compared to those who read positive articles. Although the newspaper manipulation affected private information-sharing intention, it did not affect purchase intention and public information-sharing intention. Collapsing the two conditions, participants with more negative prospects reported higher purchase intention and higher private information-sharing

intention. Together, these results suggest that negative subjective prospects about supply are associated with stronger purchase intention and private information-sharing intention.

The manipulation may not have produced larger differences in behavioral intentions because the contrast was between more and less negative prospects rather than between negative and positive prospects. Subjective prospects leaned negative across the whole sample, and even in the positive prospects condition they remained near the scale midpoint. Because the study was conducted while the 2024–2025 rice shortage was recent and rice was still expensive, the positive articles may have made participants' subjective prospects less negative rather than positive. Moreover, the distributions of subjective prospects in the two conditions overlapped considerably (Figure 2), and subjective prospects were only weakly correlated with behavioral intentions, so the difference between the conditions may have been diluted in those intentions. In addition, more participants in the positive prospects condition were excluded for failing the comprehension checks, so the two groups may have differed in attention.

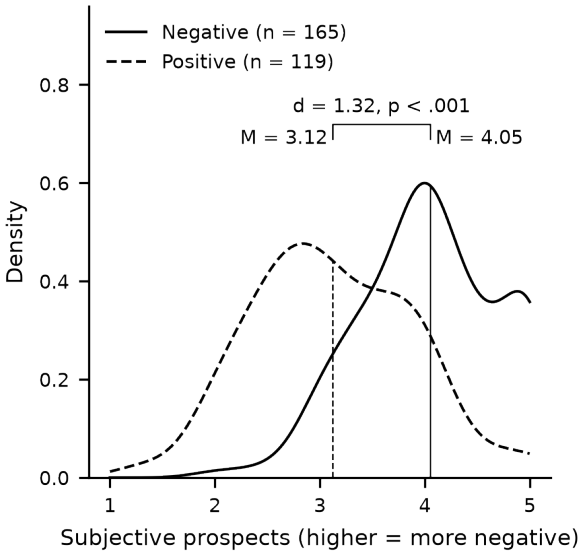


Figure 2: Distribution of subjective prospects in the negative and positive prospects conditions of Study 1 (kernel density estimates). Vertical lines mark the condition means; the bracket shows the Welch t test of the manipulation check. Higher scores indicate more negative prospects.

We conducted Study 2 to test the replicability of the correlations between negative prospects and two types of intention (i.e., purchase intention and private information-sharing intention). In addition, as the behavioral items were developed for Study 1, we examined the replicability of the three-factor structure.

4 Study 2: A Preregistered Conceptual Replication with Different Products

In Study 1, we observed significant correlations between subjective prospects and purchase/private information-sharing intention. Study 2 was a preregistered conceptual replication of Study 1 with other commodities, toilet paper and laundry detergent (OSF [OSF LINK]). The survey

was conducted in June 2026, during the 2026 Strait of Hormuz oil-supply crisis. Because a real crisis was under way, we did not manipulate subjective prospects (and we did not find their strong effects on purchase intention); instead, we measured them and assessed the correlation between subjective prospects and consumers' behavioral intentions in this crisis. In May 2026, newspaper reports named both toilet paper and laundry detergent among the daily goods likely to be affected by the oil shock [CITE: news reports, May 2026], but neither had yet run short, so participants responded to a possible shortage rather than to one they were already experiencing.

The preregistered hypotheses were:

- H1. The twelve behavioral items reproduce the three-factor structure of Study 1: purchase, private information-sharing, and public information-sharing intentions.
- H2a. Negative subjective prospects about the supply of the target commodity are positively associated with purchase intention.
- H2b. Negative subjective prospects are positively associated with private information-sharing intention.
- H3. Subjective prospects are not associated with public information-sharing intention.

The preregistration states H2a, H2b and H3 in terms of perceived stability; we report them with the reversed score, subjective prospects (where higher values mean more negative prospects), as in Study 1, thus H2a and H2b predict positive associations. The commodity was a between-participants factor included to check generalizability; no difference between commodities was hypothesized.

4.1 Method

We recruited Japanese adults aged 30 to 60, as preregistered, through Cross Marketing Inc. on June 10–11, 2026, and analyzed 536 of 1,028 responses after the preregistered exclusions (53.7% men; age $M = 44.66$, $SD = 8.71$; 257 toilet paper, 279 laundry detergent; exclusions in Appendix A.3.1). The planned sample of 500 gave power of .80 to detect $r = .13$ (significance level $\alpha = .05$, two-tailed test), slightly smaller than the smallest relevant correlation in Study 1. The study was approved by the Ethics Committee of the Graduate School of Humanities and Sociology, University of Tokyo (UTSP-25047).

Participants were randomly assigned to either toilet paper or laundry detergent condition and first read a short description of the oil-supply crisis. They then rated their subjective prospects about the supply of the commodity and responded to the same direct and indirect observation contexts as in Study 1 (with the order of the two contexts counterbalanced), shown with AI-generated images of the commodity (participants were explicitly told that the images were generated using AI; Figure 3). Six attention checks were embedded in the survey (Appendix A.3.1).

Materials of this study are available from the Open Science Framework (`https://osf.io/mgbu8/overview?view_only=93d33e1995be4b0fb09759795c3541cd`).



Figure 3: Pictures shown in the direct observation context (left) and the indirect observation context (right) of Study 2, for toilet paper (top) and laundry detergent (bottom). All pictures were generated using AI, and participants were told so.

4.1.1 Measures

Subjective Prospects. Participants' subjective prospects for the supply of the assigned commodity were measured with three 5-point scale items adapted from Study 1: whether its supply would become unstable, whether the participant could obtain it as usual, and whether it would sell out in stores. Items were coded so that higher values indicate more negative prospects and aggregated (Cronbach's $\alpha = .72$).

Purchase Intention and Information-sharing Intention. Purchase intention (three items per context) and information-sharing intention (two private sharing items and one public sharing item per context) were measured on the same 5-point scale (1 = *definitely not* to 5 = *definitely*) as in Study 1, except that the household stock levels in the purchase items were reworded to suit the target commodities (Table 5).

4.2 Results

4.2.1 Behavioral variables

To test H1, we fitted the preregistered confirmatory factor analysis (robust maximum likelihood using the lavaan package of R; R Core Team, 2026; Rosseel, 2012). The tested model imposed twelve behavioral items on three independent factors: purchase intention (six items), private information-sharing intention (four items), and public information-sharing intention (two items). We used the following criteria: CFI and TLI $\geq .90$, RMSEA and SRMR $\leq .08$.

The model did not fit the data well ($\chi^2[54] = 1103.70$, $p < .001$, CFI = .715, TLI = .652, RMSEA = .202, SRMR = .214), and the preregistered fallback allowing an oblique factor structure also failed to achieve the preregistered criteria ($\chi^2[51] = 884.59$, $p < .001$, CFI = .768, TLI = .699, RMSEA = .188, SRMR = .089). H1 was therefore not supported by the CFA. Following the next preregistered fallback, we checked the internal consistency of the three item groups; all three were acceptable ($\alpha = .88$ for purchase intention, .87 for private information-sharing intention, and .86 for public information-sharing intention; $r = .75$ between the two public information-sharing items), so we used their item means to test H2a, H2b and H3.

An exploratory factor analysis (minimum residual extraction with varimax rotation, as in Study 1) recovered the three factors observed in Study 1 (Table 2). Three eigenvalues exceeded 1 (5.03, 2.40 and 1.21; the fourth was 0.83), and the three factors explained 63.4% of the total variance. Every item loaded highest on its preregistered factor, and the promax solution gave the same pattern (Appendix A.3.2).

4.2.2 Subjective prospects and behavioral intentions

We tested H2a, H2b and H3 by computing Pearson's product-moment correlations between subjective prospects and behavioral intention scores in the pooled sample (Table 3). The subjective prospects × purchase intention correlation was significant ($r[534] = .18$, 95% CI [.09, .26], $p < .001$), supporting H2a, and the subjective prospects × private information-sharing intention correlation was also significant ($r[534] = .20$, 95% CI [.11, .28], $p < .001$), supporting H2b. Contrary to H3, the subjective prospects × public information-sharing intention correlation was

Table 2: Varimax-rotated loadings of the twelve behavioral items in Study 2 (toilet paper and laundry detergent, $N = 536$). Minimum residual extraction; loadings $\geq .40$ in bold. Full item wording is given in Table 5.

Item	Context	Purchase	Private	Public
Buy: a few days left	Direct	**.67**	.05	−.21
Buy: half left	Direct	**.81**	.13	−.02
Buy: usual left	Direct	**.76**	.16	.08
Go to buy: a few days	Indirect	**.67**	.23	−.06
Go to buy: half	Indirect	**.75**	.28	.15
Go to buy: usual	Indirect	**.68**	.31	.21
Private sharing: close	Direct	.25	**.72**	.02
Private sharing: casual	Direct	.13	**.76**	.28
Private sharing: close	Indirect	.27	**.78**	.08
Private sharing: casual	Indirect	.18	**.73**	.32
Public sharing	Direct	−.02	.21	**.76**
Public sharing	Indirect	.01	.18	**.90**
% variance		27.7	21.6	14.2

also significant ($r[534] = .19$, 95% CI [.10, .27], $p < .001$). These associations held for both commodities when each commodity was analyzed separately (Appendix A.3.3).

4.3 Discussion

Across the two studies, consumers with more negative subjective prospects responded to shortage information with higher purchase intention and higher private information-sharing intention, as hypothesized, with correlations of about .2 in both studies (Table 3 and Figure 4). Responses in different contexts (i.e., direct and indirect observation) loaded on the same factor as long as their behavioral outputs were the same (purchase vs. privately sharing information vs. publicly sharing information). Although CFA failed to support this three-factor structure with strict criteria, EFA replicated it. Moreover, in both Studies 1 and 2, negative prospects for supply were positively associated with purchase intention and private information-sharing intention.

Table 3: Means, standard deviations and Pearson correlations of subjective prospects and the behavioral intentions in Study 1 (rice, $N = 284$) and Study 2 (toilet paper and laundry detergent, pooled, $N = 536$). $* \ p < .05$, $** \ p < .01$, $*** \ p < .001$.

Variable	*M*	*SD*	1	2	3
Study 1: Rice ($N = 284$)					
1. Subjective prospects	3.66	0.84	–		
2. Purchase intention	2.77	0.81	.18**	–	
3. Private information-sharing intention	2.83	0.87	.18**	.49***	–
4. Public information-sharing intention	1.71	0.81	.05	.05	.30***
Study 2: Toilet paper and detergent ($N = 536$)					
1. Subjective prospects	2.96	0.66	–		
2. Purchase intention	3.44	0.84	.18***	–	
3. Private information-sharing intention	2.94	0.94	.20***	.45***	–
4. Public information-sharing intention	1.67	0.83	.19***	.07	.36***

In Study 2, the preregistered CFA did not support the hypothesized three-factor structure.

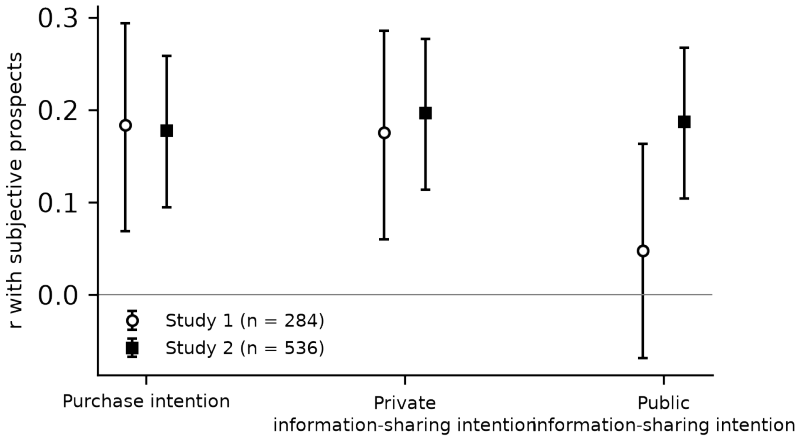


Figure 4: Correlations of subjective prospects with the three behavioral intentions in Study 1 (rice, $N = 284$) and Study 2 (toilet paper and laundry detergent, $N = 536$), with 95% confidence intervals.

However, EFA yielded the hypothesized three-factor structure: all factors were loaded by exactly the same items as in Study 1. In retrospect, the preregistered CFA criteria might have been too strict because we fitted a non-theoretically derived model, which was used to empirically organize different behavioral responses in different contexts.

In both studies, the shortage information was the stimulus: participants saw a context with a nearly empty shelf or an SNS post and reported how they would respond. Subjective prospects, which indicate how strongly a consumer perceives a crisis, were linked to how strongly they responded to this stimulus, but only by a small amount. Whether such small individual changes can produce a shortage in the market as a whole cannot be answered by surveys; the next section addresses it with a simulation in which shortage information triggers purchase and information sharing.

5 Simulation: Shortage Information on a Network of Close Relationships

In Studies 1 and 2, negative subjective prospects about future supply were consistently associated with higher purchase intention and private information-sharing intention, whereas public information-sharing intention remained rare and inconsistently linked to subjective prospects. However, these effects were small: consumers with more negative subjective prospects intended to buy and share only slightly more, consistent with earlier findings that most consumers react to rumors and crises with modest rather than extreme behavioral changes [CITE]. Yet real shortages often involve severe depletion that spreads widely within a short period [CITE]. Because survey data capture individual intentions at a single point in time, they cannot show how small individual shifts combine into such market-level outcomes. This chapter uses a simulation to test whether small increases in purchase intention and private information-sharing intention, interacting with fixed local store stock and a social network of mostly local ties and a few long-range ties, are enough to produce rapid, widespread depletion without assuming hoarding or dramatic changes in individual behavior.

5.1 Model

To represent a market during a brief crisis window before retailers can restock, the model places local retailers with fixed, non-replenished stock and consumers on a spatial ring where shoppers visit their home district first and search only nearby districts if their local store is empty. Because total market supply is fixed over this short window, the central market-level outcome is how quickly the fixed supply runs out (depletion time). Reflecting the predominance of private over public information sharing observed in Studies 1 and 2, the model omits market-wide broadcasting and restricts information transmission to a small, finite number of undirected close ties per consumer, along which shortage information can flow both ways. Although shopping is physically constrained to a consumer's local neighborhood, close contacts such as parents or hometown friends often live in other regions served by different local retailers and communicate by phone or private messaging; we therefore model the communication structure as a small-world network (Watts & Strogatz, 1998) consisting of mostly local neighborhood ties and a small fraction of rewired long-range ties (p_{rewire}).

In the model, consumers make the two decisions measured in Studies 1 and 2: when to shop and whether to share shortage information. Each consumer shops at most once, buying a single unit, and shares at most twice, with one opportunity for sharing after indirect observation before shopping and one for sharing after direct observation upon finding an empty store. In Studies 1 and 2, consumers with more negative subjective prospects reported higher purchase intention when they encountered shortage information, whether directly at a depleted shelf or indirectly through a shared message. In the model, shortage messages received through close ties likewise act as cues that raise a consumer's shopping probability. How easily they do so is set by reaction sensitivity (S), which ranges from 0 to 1: at $S = 1$, a message from a single close tie is enough to trigger the full shopping reaction, whereas at $S = 0$, a consumer stays near the baseline shopping probability even when all of their close ties share. Consumers can share shortage information acquired from two sources, matching the two observation contexts in Studies 1 and 2: sharing after direct observation, when they encounter an empty store themselves during a shopping trip, and sharing after indirect observation, when they forward shortage information received through a close tie. In Studies 1 and 2, sharing after direct observation and sharing after indirect observation loaded on a single factor, private information-sharing intention, and correlated strongly ($r = .68$ in Study 1 and .75 in Study 2); sharing after indirect observation was only slightly lower in mean (87% of sharing after direct observation in Study 1 and 93% in Study 2 on a rescaled 0–1 range; Appendix). The model therefore governs both with a single private sharing probability (p_{close}) and fixes sharing after indirect observation at a constant fraction ($0.8 \times p_{\text{close}}$) of sharing after direct observation. The model leaves out price adjustments, utility maximization, and individual stockpiling, so it tests whether rapid market-wide depletion can emerge from earlier shopping and information sharing alone.

Concretely, the market is initialized with $N = 2,500$ homogeneous consumers distributed evenly across $D = 25$ districts on a circular ring (100 residents per district), where consumer c resides in district $\lfloor cD/N \rfloor$ (Figure **??**; Table 4). At an abundance ratio of .50, each retailer holds 50 units of non-replenished stock, except for district 0, which is initialized with 0 units at step 0 (leaving 1,200 units across the market) to seed a single local shortage without market-

wide news. Each consumer is assigned a fixed five-district shopping route starting at their home district followed by the two adjacent districts on each side in a randomized order, and starts with $k = 10$ local close ties (the five nearest consumers on each side on the ring, almost all of whom live in the same district) before each tie is rewired to a randomly chosen consumer with probability p_{rewire}. When $I_{i,t}$ of consumer i's close ties are actively sharing at step t, the consumer's shopping probability is

$$P_{i,t}(\text{shop}) = \begin{cases} P_0, & I_{i,t} = 0, \\ P_0 + \dfrac{P_{\max} - P_0}{1 + e^{-G(I_{i,t} - M(S))}}, & I_{i,t} > 0, \end{cases} \qquad M(S) = 0.5 + k(1 - S),$$

where $P_0 = .01$ is the baseline rate, $P_{\max} = .10$ is the maximum rate, $G = 20$ is the steepness, and $M(S)$ is the reaction midpoint set by reaction sensitivity $S \in [0, 1]$. At $S = 1$, $M = 0.5$, so a single sharing tie raises the shopping probability to nearly $P_{\max}$; at $S = 0$, $M = 10.5$, so even $k = 10$ sharing ties leave it near P_0. Because $I_{i,t}$ is an integer count and $M(S)$ always falls halfway between two integers, this steep sigmoid effectively acts as a step function: the shopping probability jumps from near P_0 to near $P_{\max}$ once $I_{i,t}$ exceeds $M(S)$. We use this sharp threshold deliberately for simplicity; a smoother curve (lower G) yields similar results (Appendix A.4.2).

[PLACEHOLDER: Reworked conceptual diagram (`thesis/assets/sim/network-concept.png` / `results/figures/sim_concept.svg`). Illustrates spatial districts on the ring, local vs. rewired social ties ($k = 10$, rewiring probability p_{rewire}), the 5-district local shopping route, and the single-consumer state transitions for shopping once and sharing firsthand (p_{close}) or secondhand ($0.8 \times p_{\text{close}}$).]

Table 4: Fixed and manipulated parameters in the network simulation model.

Parameter	Value
Consumers (N)	2,500
Districts (D)	25 (100 residents per district)
Initial stock per district	50 units (abundance = .50; 1,200 units total)
Seeded initial shortage	District 0 starts with 0 units at step 0
Shopping route per consumer	Home district + 2 adjacent districts on each side (5
Initial network degree (k)	10 close ties (5 on each side on the consumer ring)
Network rewiring probability (p_{rewire})	Manipulated (e.g., 0% to 10%)
Close-sharing probability (p_{close})	Manipulated (0 to 1.0)
Secondhand forwarding probability	$0.8 \times p_{\text{close}}$
Sharing duration	10 steps
Baseline / maximum shopping probability (P_0 / $P_{\max}$)	.01 / .10 per step
Sigmoid steepness (G)	20
Reaction sensitivity (S) / reaction midpoint (M)	Manipulated ($S \in [0, 1]$, step .1; $M(S) = 0.5 + k(1$
Depletion criterion	All 24 initially stocked districts empty
Maximum steps	150 steps
Replicates per condition	40 matched random seeds

Each simulation run proceeds in discrete time steps, and each step applies five stages in a fixed order to all consumers at once: (1) consumers who decided to share in the previous

step start sharing for a fixed duration of 10 steps, and each consumer counts the number of close ties currently sharing, $I_{i,t}$; (2) consumers who hear of the shortage ($I_{i,t} \geq 1$) for the first time and have not yet committed to shopping get a single chance for sharing after indirect observation, with probability $0.8 \times p_{\text{close}}$; (3) consumers who have not yet committed to shopping update their shopping probability $P_{i,t}(\text{shop})$ and, if they decide to shop, commit permanently to shopping starting at their home district; (4) each district allocates its remaining stock at random among that step's visitors, where successful buyers purchase one unit and exit the market while unsuccessful shoppers find an empty store, get a single chance for sharing after direct observation, with private sharing probability p_{close}, and move to the next district on their five-district route for the following step (giving up once all five districts on their route are empty); and (5) consumers who have shared for 10 steps stop sharing. The primary outcome is *depletion time* T, defined as the first step at which all 24 initially stocked districts have run out of stock (up to a ceiling of 150 steps), averaged across 40 matched random seeds per parameter condition, with independent random streams for initialization, network generation, shopping, sharing, and stock allocation.

5.2 Results

[NOTE: The colors and layout of the simulation figures are not final yet; the design is still being refined.]

To establish the range of possible depletion times, we first set the private sharing probability to zero ($p_{\text{close}} = 0$) so that every consumer had the same constant shopping probability, and varied this probability from $P_0 = .01$ to $P_{\max} = .10$. We chose these two values for illustration rather than estimating them from data, so the simulation shows qualitative patterns, not quantitative predictions. As in all runs, district 0 started with no stock. At P_0, every consumer shopped at the baseline probability, as in a normal market with no crisis and no panic buying. The market ran out of stock after 73.8 steps (the slow bound), which serves as the reference point for normal shopping. At $P_{\max}$, where every consumer shopped at the maximum shopping probability from the first step, the market ran out of stock after 8.3 steps (the fast bound). Depletion time fell steadily between these bounds. We use this range to interpret all later results: because shortage news affects the market only through the shopping probability, a depletion time near the slow bound means that the news barely increased shopping, whereas one near the fast bound means that it pushed almost all consumers to the maximum shopping probability.

[QUESTION: Is a figure of depletion time across the full range of shopping probabilities useful here, or is the range in the text enough?]

Next, we varied reaction sensitivity (S from 0 to 1) and private sharing probability (p_{close} from 0 to 1) on a network without long-range ties ($p_{\text{rewire}} = 0$; Figure 5, first panel). Depletion time remained near the slow bound across almost all parameter combinations; it was slightly shorter (12% to 15% faster than the slow bound) only when both reaction sensitivity and private sharing probability were near their maximum. Even when a single sharing tie was enough to make consumers react and every consumer shared what they saw or heard, shortage news stayed near district 0, and the rest of the market ran out of stock at the normal rate.

When a small share of close ties was rewired into long-range ties ($p_{\text{rewire}} > 0$), however, the

two behavioral parameters together produced a shift from slow to fast depletion (Figure 5). In each panel, reaction sensitivity increases upward and private sharing probability increases to the right. Even at a low rewiring probability (0.5%), a region of fast depletion opened in the upper-right corner. The boundary between the slow and fast regions was gradual at low rewiring and became sharper as rewiring increased (2% and 10%). Long-range ties allowed shortage news from district 0 to reach distant districts, where consumers started shopping and moved on to neighboring districts when they found empty stores. The effect of rewiring saturated quickly: beyond 5% rewiring, the panels changed little, so a very small rewiring probability was enough to produce the strong contrast between the upper-right and lower-left corners.

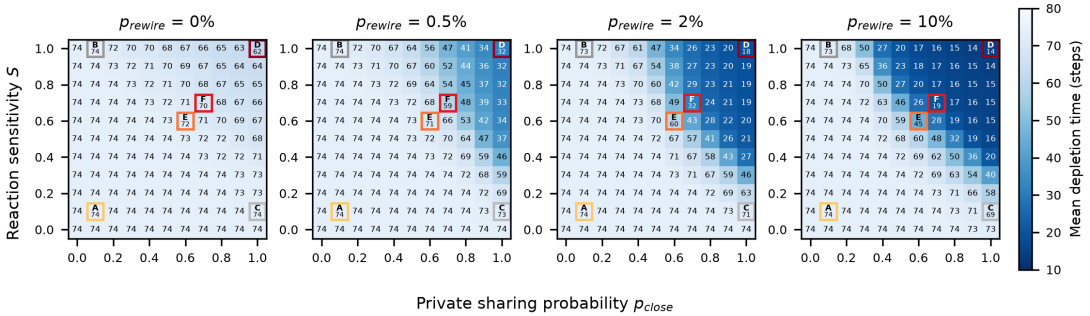


Figure 5: Mean depletion time (steps) across reaction sensitivity (S) and private sharing probability (p_{close}) at selected rewiring probabilities (p_{rewire}). Reaction sensitivity increases upward and private sharing probability increases to the right. Each cell shows the mean of 40 matched runs. The full set of rewiring probabilities is shown in Appendix A.4.1.

To illustrate how individual behavioral combinations behaved across the full range of rewiring probabilities, Figure 6 plots mean depletion time as a function of rewiring probability (p_{rewire}, starting from 0%) for six representative parameter pairs: the four extreme corner regimes (low sensitivity–low sharing, high sensitivity–low sharing, low sensitivity–high sharing, and high sensitivity–high sharing) and two closely spaced parameter pairs located on either side of the boundary (corners: $S = .1$ or 1 and $p_{\text{close}} = .1$ or 1; near-boundary pairs: $(S, p_{\text{close}}) = (.6, .6)$ vs. $(.7, .7)$). At $p_{\text{rewire}} = 0\%$, five of the six pairs remained near the slow bound; only the high–high pair was somewhat faster. As rewiring increased, the low–low and high-sensitivity–low-sharing pairs remained near the slow bound, the low-sensitivity–high-sharing pair became only slightly faster, and the high–high pair fell sharply by 1% rewiring before approaching the fast bound at high rewiring. Once rewiring was present, a small increase in both behavioral parameters from the pair just below the boundary ($S = .6$ and $p_{\text{close}} = .6$) to the pair just above it ($S = .7$ and $p_{\text{close}} = .7$) produced a large drop in depletion time, showing that a small change in behavior could have a massive effect. Without rewiring, the two near-boundary lines were almost together, but with rewiring they separated widely, and the width of the gap between them in Figure 6 showed how much the risk of market-wide depletion increased.

Together, the heatmaps (Figure 5) and the line plot (Figure 6) show the same pattern. When consumers shopped near home and a small share of their close ties were rewired into long-range ties, each heatmap panel split into a region where the market ran out of stock at about the normal rate and a region where it ran out quickly. In the line plot, the two parameter pairs on either side of this boundary differed only slightly in reaction sensitivity and private sharing

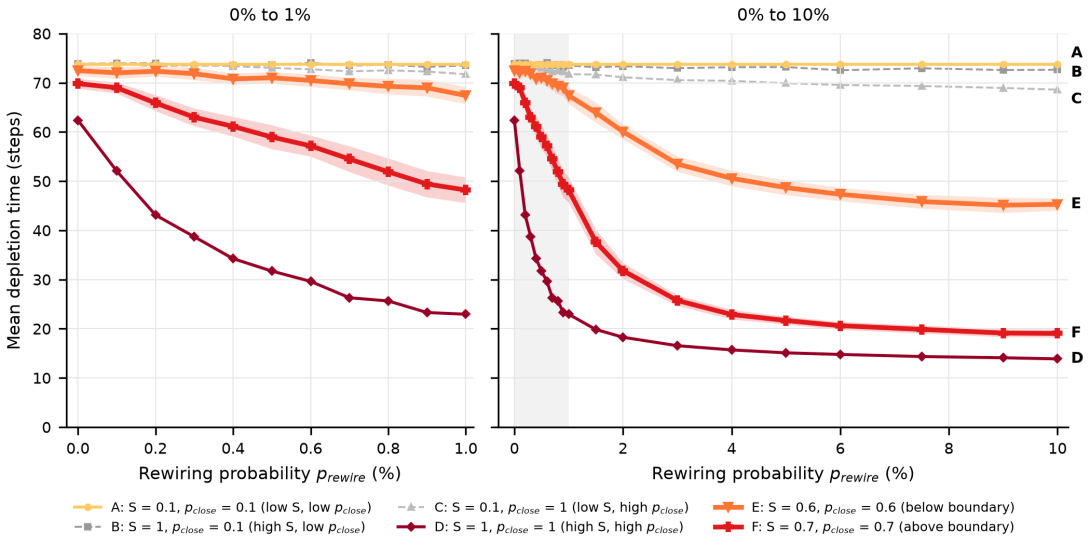


Figure 6: Mean depletion time (steps) across rewiring probability (p_{rewire}, starting from 0%) for four corner regimes (low–low, high–low, low–high, high–high) and two closely spaced combinations straddling the boundary. Each point is the mean of 40 matched runs.

probability, yet their depletion times were far apart. Although the effects on purchase intention and private sharing in Studies 1 and 2 were modest, this result demonstrates that a slight joint increase in reaction sensitivity and private sharing probability can still, through interacting consumers, lead to fast market-wide depletion, as long as a few close ties are rewired to reach other districts.

6 General Discussion

This thesis asked how panic buying can empty a whole market in a few days when the behavior of each consumer changes only a little. Two surveys measured how consumers respond to shortage information, and an agent-based model asked what these responses imply for a market.

6.1 Small individual differences

At the individual level, the effects were consistently small. In Study 1, news about the future supply of rice changed subjective prospects by more than one standard deviation but changed purchase and information-sharing intentions little or not at all. In both studies, participants with more negative subjective prospects had higher purchase intention and higher private information-sharing intention, with correlations of about .18–.20 (Table 3). These correlations were similar across rice, toilet paper and laundry detergent, across the positive and negative prospects conditions, and with and without the crisis context. In Study 1, the correlations did not differ detectably between the news conditions, even though the conditions differed strongly in subjective prospects. The link between subjective prospects and behavior was therefore stable, and it was weak: subjective prospects accounted for about 3–4% of the variance in purchase intention and private information-sharing intention.

The structure of the responses was the most stable result. In both studies, responses separated into purchase intention, private information-sharing intention, and public information-

sharing intention, and not by whether the shortage was observed directly on a shelf or indirectly in an SNS post. Public information-sharing intention was low in both studies. Purchase intention and public information-sharing intention were unrelated, and private information-sharing intention was related to both. The preregistered CFA with independent factors failed, but the EFA recovered the same three groups of items in both studies, so the three types of response are better seen as distinct but correlated.

The role of public information-sharing intention was the main inconsistency. Subjective prospects were unrelated to public information-sharing intention for rice ($r = .05$) but related to it for toilet paper and detergent during the oil-supply crisis ($r = .19$). Whatever its size, the association concerns a behavior that few participants said they would perform. The evidence therefore does not support the idea that consumers with negative prospects drive panic buying mainly by broadcasting shortages to strangers. It is more consistent with shortage information spreading through private information sharing among close relationships.

6.2 From individuals to the market

The simulation shows why small individual differences do not imply small aggregate effects. When private sharing and sensitivity to shared information were both set to modest values, the market behaved as if there were no information at all. Raising either one alone had a partial effect. Raising both made depletion about 70% faster, far more than the sum of the separate effects. Most of this acceleration required only a small share of close relationships that reach other districts. In such a market, small shifts in two behaviors, each hard to detect in a survey, can together turn a local shortage into a market-wide one.

This suggests a view of what changes in an emergency. Study 1 found no sign that negative prospects news changes how strongly subjective prospects are linked to behavior. Study 2 found that concern about the crisis added to purchase intention and private information-sharing intention without amplifying the effect of subjective prospects. An emergency may therefore work less by changing how each person responds and more by shifting many people slightly in the same direction at the same time: slightly more negative prospects, slightly higher willingness to share privately with family and friends, and a slightly quicker shopping response when they hear about a shortage. Common sources such as mass media reports could produce such simultaneous shifts [CITE]. Because the market can sit close to a joint threshold, these shifts do not need to be large.

This view also explains why panic buying is rare despite the constant availability of information channels. In the model, the conservative setting stayed near the baseline across the whole range of network structures. The same networks that carry a panic in an emergency carry little in normal times, because too few people share and too few react.

6.3 Limitations

The surveys measured intentions in hypothetical observation contexts, not actual purchases or messages, and both used online panels. Many responses were excluded for failed comprehension or attention checks (46% of consenting responses in Study 1 and 41% of eligible responses in Study 2), so the samples are selected toward attentive participants. In Study 1 the comprehen-

sion checks excluded more participants in the positive prospects condition, which weakens the comparison between conditions. The correlations are cross-sectional and do not show that subjective prospects cause purchase or sharing. The simulation is not calibrated: its parameters are chosen to explore the model, and the surveys constrain only its structure. The model assumes homogeneous consumers, a fixed network, no restocking, and no public information sharing.

6.4 Future directions

Three steps would connect the parts of this thesis more closely. First, the distributions of information-sharing and purchase intentions in the surveys could inform heterogeneous consumers in the model, instead of one value for everyone. Second, the number of close contacts who live and shop in other areas could be measured directly, since the simulation shows that this share decides whether a local shortage can spread. Third, observed purchase data or messaging data from a real shortage would show whether the small differences in intentions found here appear in behavior.

6.5 Conclusion

Individual consumers respond to shortage information in modest and stable ways. They report higher purchase intention and private information-sharing intention when they hold more negative subjective prospects about supply, and they rarely broadcast shortages publicly. The simulation shows that such modest responses are enough for a market-wide shortage when several of them shift together and when close relationships reach beyond the local neighborhood. Large-scale panic buying does not require panicking consumers; it can emerge from small, simultaneous changes in ordinary behavior.

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A Appendix

Placeholder, to be completed. Details moved out of the main text are collected here.

A.1 Behavioral items

A.2 Study 1

A.2.1 Exclusions

Responses were excluded in the order shown in Table 6. We dropped every response whose ID or IP address appeared more than once because we could not tell which response was valid (6 cases). We excluded 39 participants who did not agree to participate; they were redirected to the end screen and gave no responses. Based on responses to comprehension checks, we excluded 246 participants with at least one wrong answer to the four items: 100 from the negative prospects condition and 146 from the positive prospects condition. We also inspected response duration. The shortest response took 70 seconds; because the survey was short and these participants passed all comprehension checks, no response was excluded due to short response duration.

Table 5: Behavioral items in Studies 1 and 2 (translated from Japanese). Each item was answered on a 5-point scale from *definitely not* to *definitely*. Target is the factor found in the Study 1 EFA and preregistered for Study 2.

Context	Target	Study 1 (rice)	Study 2 (toilet paper, laundry detergent)
Direct	Purchase	Buy the remaining rice; a few days of rice left at home	Buy the remaining stock; only a few days' worth at home
Direct	Purchase	Same; more than a week of rice left	Same; about half the usual amount at home
Direct	Purchase	Same; more than a month of rice left	Same; the usual amount at home
Direct	Private sharing	Tell housemates (family), best friends or a romantic partner	Tell housemates (family), best friends or a romantic partner
Direct	Private sharing	Tell acquaintances or friends	Tell acquaintances or friends
Direct	Public sharing	Post to many unspecified people on SNS	Post to many people on SNS
Indirect	Purchase	Go to buy rice right away; a few days of rice left	Go to buy right away; only a few days' worth
Indirect	Purchase	Same; more than a week of rice left	Same; about half the usual amount
Indirect	Purchase	Same; more than a month of rice left	Same; the usual amount
Indirect	Private sharing	Tell housemates (family), best friends or a romantic partner	Tell housemates (family), best friends or a romantic partner
Indirect	Private sharing	Tell acquaintances or friends	Tell acquaintances or friends
Indirect	Public sharing	Share or post to many unspecified people	Repost or share to many people

Table 6: Study 1: responses excluded at each step.

Step	Excluded	Remaining
Raw responses		575
Unfinished	0	575
Participant ID or IP address repeated	6	569
No consent	39	530
Any wrong comprehension answer	246	284

A.2.2 Household questions

Before reading the articles, participants reported their household size, whether they were the main buyer of food in the household, how often per week they cooked rice and ate rice, and how much rice they cooked at a time. [PLACEHOLDER: descriptive statistics, and the condition comparisons and correlations repeated with these variables as controls.]

A.2.3 Behavioral variables

Expected structure. We had designed the items expecting the observation context, direct or indirect observation, to separate the items, and the sharing items to form one factor. Neither happened: purchase items from both contexts loaded on one factor, and sharing split into private and public information-sharing intentions.

Promax rotation. The promax solution gave the same three factors as the varimax solution. The factor correlations were .56 between purchase intention and private information-sharing intention, .18 between private and public information-sharing intention, and −.01 between purchase intention and public information-sharing intention. Table 7 shows the promax loadings.

Table 7: Promax-rotated loadings of the twelve behavioral items in Study 1 (rice, $N = 284$). Minimum residual extraction; loadings $\geq .40$ in bold.

Item	Context	Purchase	Private	Public
Buy: a few days left	Direct	**.48**	.19	−.30
Buy: a week left	Direct	**.80**	.03	−.21
Buy: a month left	Direct	**.82**	−.14	.15
Go to buy: a few days	Indirect	**.58**	.21	−.08
Go to buy: a week	Indirect	**.83**	.00	.03
Go to buy: a month	Indirect	**.82**	−.15	.29
Private sharing: close	Direct	−.06	**.87**	−.14
Private sharing: casual	Direct	−.13	**.82**	.17
Private sharing: close	Indirect	.15	**.67**	.05
Private sharing: casual	Indirect	.08	**.65**	.29
Public sharing	Direct	−.01	.11	**.79**
Public sharing	Indirect	.04	.06	**.85**

Regression factor scores. We repeated the condition comparisons and the correlations with regression factor scores from the varimax solution instead of item means. Private information-sharing intention was again slightly higher in the negative prospects condition ($t[257] = 2.15$, $p = .032$, $d = 0.26$); purchase intention ($p = .760$) and public information-sharing intention ($p = .977$) did not differ. Subjective prospects correlated with purchase intention ($r = .16$, $p = .006$) and private information-sharing intention ($r = .15$, $p = .014$) but not with public information-sharing intention ($r = .03$, $p = .589$).

Table 8: Study 1: subjective prospects and behavioral intentions by news condition, with Welch's t-tests (negative minus positive). Values are M (SD).

Measure	Negative ($n = 165$)	Positive ($n = 119$)	t	df	p	d
Subjective prospects	4.05 (0.66)	3.12 (0.76)	10.75	231	$< .001$	1.32
Purchase intention	2.79 (0.83)	2.73 (0.78)	0.64	263	.521	0.08
Private information-sharing intention	2.91 (0.88)	2.70 (0.85)	2.02	258	.044	0.24
Public information-sharing intention	1.72 (0.85)	1.70 (0.75)	0.18	270	.853	0.02

Table 9: Study 1: means, standard deviations and Pearson correlations of subjective prospects and the behavioral intentions, pooled over conditions ($N = 284$). 95% confidence intervals in brackets. $* \; p < .05$, $** \; p < .01$, $*** \; p < .001$.

Variable	M	SD	1	2	3
1. Subjective prospects	3.66	0.84			
2. Purchase intention	2.77	0.81	.18** [.07, .29]		
3. Private information-sharing intention	2.83	0.87	.18** [.06, .29]	.49*** [.39, .57]	
4. Public information-sharing intention	1.71	0.81	.05 [−.07, .16]	.05 [−.07, .16]	.30*** [.19, .41]

Table 10: Study 1: correlations between subjective prospects and the behavioral intentions within each news condition, compared with Fisher's r-to-z test.

Outcome	r (negative, $n = 165$)	r (positive, $n = 119$)	z	p
Purchase intention	.26	.12	1.15	.249
Private information-sharing intention	.18	.08	0.84	.398
Public information-sharing intention	.05	.05	0.00	$> .999$

A.2.4 Effect of the news condition

A.2.5 Correlations among the variables

A.2.6 Correlations within each condition

A.2.7 Sharing after direct and indirect observation

Averaging the two items about private information-sharing intention within each observation context, participants were more willing to share a shortage they had observed directly ($M = 2.96$, $SD = 0.95$) than one they had observed indirectly in an SNS post ($M = 2.69$, $SD = 0.95$; $t[283] = 5.85$, $p < .001$, $d_z = 0.35$). The two composites were strongly correlated ($r = .68$). After rescaling both means to 0–1, sharing after indirect observation was 87% of sharing after direct observation. The difference did not depend on the condition: it was 0.32 in the negative prospects condition and 0.19 in the positive prospects condition (Welch's $t[262] = 1.43$, $p = .155$, $d = 0.17$).

A.3 Study 2

A.3.1 Exclusions

During data collection, Qualtrics did not count responses with a wrong answer in the attention checks toward the planned sample of 500. As preregistered, we closed the survey once the counted responses reached 500, and participants who had already started were allowed to finish, which gave 1,028 raw responses. Responses were excluded by the preregistered rules, in the order shown in Table 11. Six attention checks were embedded in the observation context blocks and at the end of the survey: three asked whether the participant had looked at the image, two asked for a specified answer, and one asked whether the participant had read the questions carefully. A last check for a repeated participant ID or IP address, not in the preregistration, excluded no response.

Table 11: Study 2: responses excluded at each step.

Step	Excluded	Remaining
Raw responses		1028
Unfinished	0	1028
No consent	80	948
Age outside 30–60	44	904
Duration under 60 s	3	901
Any failed attention check	365	536
Missing analysis variables	0	536
Participant ID or IP address repeated	0	536

A.3.2 Robustness checks

Number of factors. As in Study 1, we kept the three factors with eigenvalues greater than 1; the fourth and fifth eigenvalues of the item correlation matrix were 0.83 and 0.69.

Promax rotation. Every item loaded highest on its preregistered factor in both the varimax solution (purchase intention .67–.81, private information-sharing intention .72–.78, public information-sharing intention .76 and .90) and the promax solution. The promax factor correlations were .48 between purchase intention and private information-sharing intention, .44 between private and public information-sharing intention, and .07 between purchase intention and public information-sharing intention. [PLACEHOLDER: promax loading table.]

A.3.3 Exploratory analyses

Commodity. Subjective prospects were more negative for laundry detergent ($M = 3.06$, $SD = 0.60$) than for toilet paper ($M = 2.85$, $SD = 0.70$; $t[504] = 3.71$, $p < .001$, $d = 0.32$). Purchase intention was higher for toilet paper ($M = 3.56$, $SD = 0.80$) than for laundry detergent ($M = 3.33$, $SD = 0.87$; $t[534] = 3.23$, $p = .001$, $d = 0.28$), while private information-sharing intention, public information-sharing intention, and crisis concern did not differ ($p \geq .114$). Because subjective prospects differed between commodities, the preregistered exploratory plan called for commodity-specific correlations. They were similar for both commodities: for toilet paper ($n = 257$), $r = .21$ (purchase intention), .19 (private information-sharing intention), and .16 (public information-sharing intention); for laundry detergent ($n = 279$), $r = .20$, .21, and .24 (all $p \leq .009$).

Crisis context. Three exploratory items asked how concerned participants were about the oil-supply situation, how much it would affect their daily shopping, and how long it would last (5-point scales; $M = 3.91$, $SD = 0.84$, $\alpha = .67$). Concern about the crisis was related to subjective prospects about the commodity ($r = .17$, $p < .001$), to purchase intention ($r = .26$, $p < .001$), and to private information-sharing intention ($r = .18$, $p < .001$), but not to public information-sharing intention ($r = -.05$, $p = .255$). In regressions with commodity, subjective prospects, crisis context and their interaction (predictors mean-centered), both subjective prospects and crisis context predicted purchase intention ($b = 0.21$ and 0.24, both $p < .001$) and private information-sharing intention ($b = 0.25$ and 0.18, both $p < .001$). For public information-sharing intention, only subjective prospects did ($b = 0.27$, $p < .001$; crisis $b = -0.08$, $p = .061$). The interaction was not significant for any outcome (all $p \geq .222$).

A.3.4 Sharing after direct and indirect observation

Averaging the two items about private information-sharing intention within each observation context, participants were more willing to share a shortage they had observed directly ($M = 3.01$, $SD = 1.00$) than one they had observed indirectly in an SNS post ($M = 2.87$, $SD = 1.01$; $t[535] = 4.47$, $p < .001$, $d_z = 0.19$). The two composites were strongly correlated ($r = .75$). After rescaling both means to 0–1, sharing after indirect observation was 93% of sharing after direct observation. [PLACEHOLDER: test whether the difference between direct and indirect observation depends on subjective prospects.]

A.4 Simulation

A.4.1 Depletion time at all rewiring probabilities

Figure 7 shows mean depletion time for every rewiring probability we ran, from 0% to 1% in steps of 0.1% and at 2%, 5%, and 10%. The region of fast depletion first appeared at 0.2% and grew steadily with rewiring.

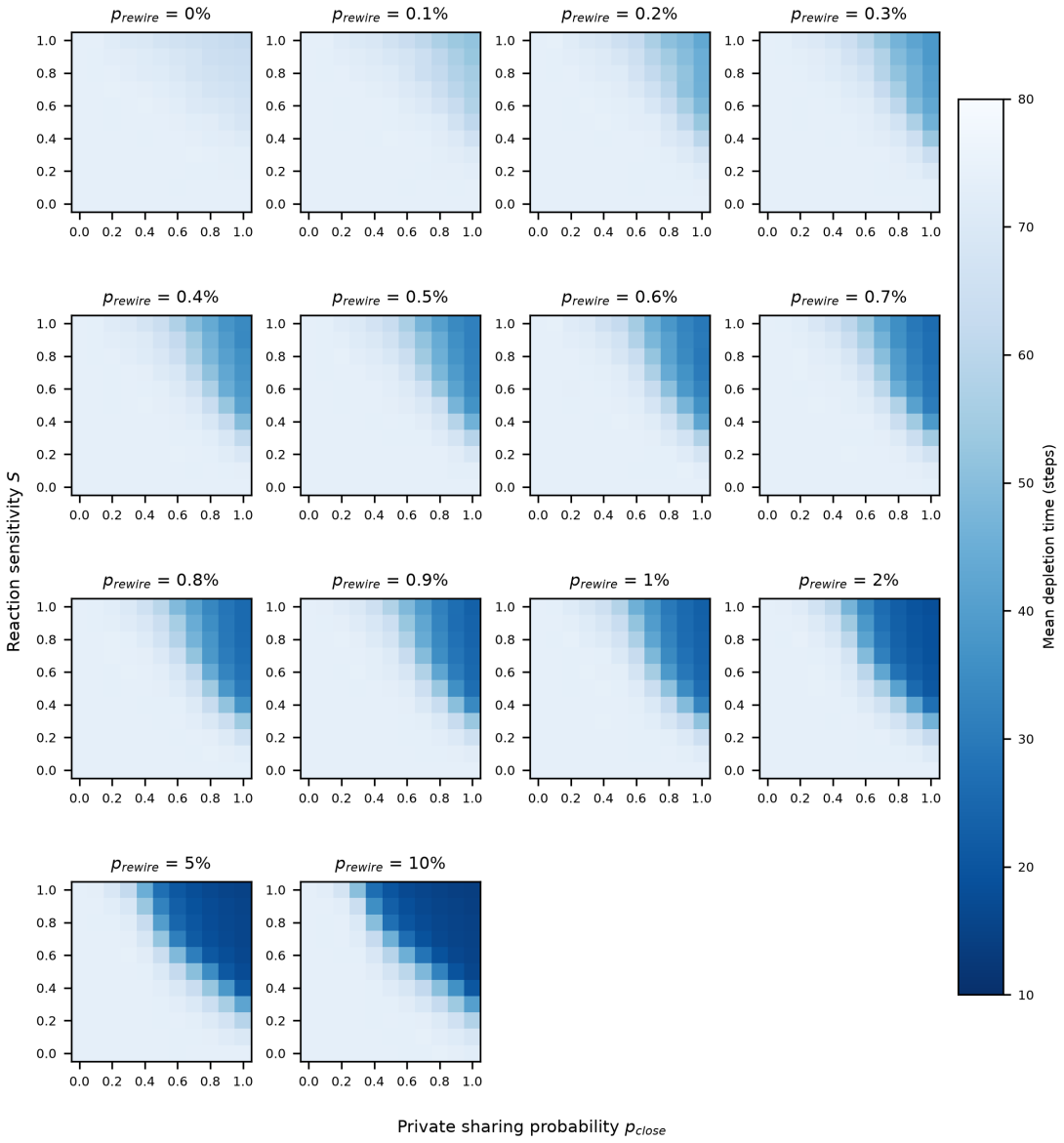


Figure 7: Mean depletion time (steps) across reaction sensitivity (S) and private sharing probability (p_{close}) at all rewiring probabilities (p_{rewire}). Same color scale as Figure 5. Each cell shows the mean of 40 matched runs.

A.4.2 Smoother reaction curve

To check that the main results do not depend on the step-like reaction curve ($G = 20$), we repeated the heatmaps with a smooth curve ($G = 1$ and $G = 2$). The results were almost the same (Figure 8): without rewiring there was no fast region, any rewiring opened a fast

region in the upper-right corner, and higher rewiring shifted the boundary toward lower reaction sensitivity and private sharing probability. The only visible difference was a slightly wider boundary at $G = 1$.

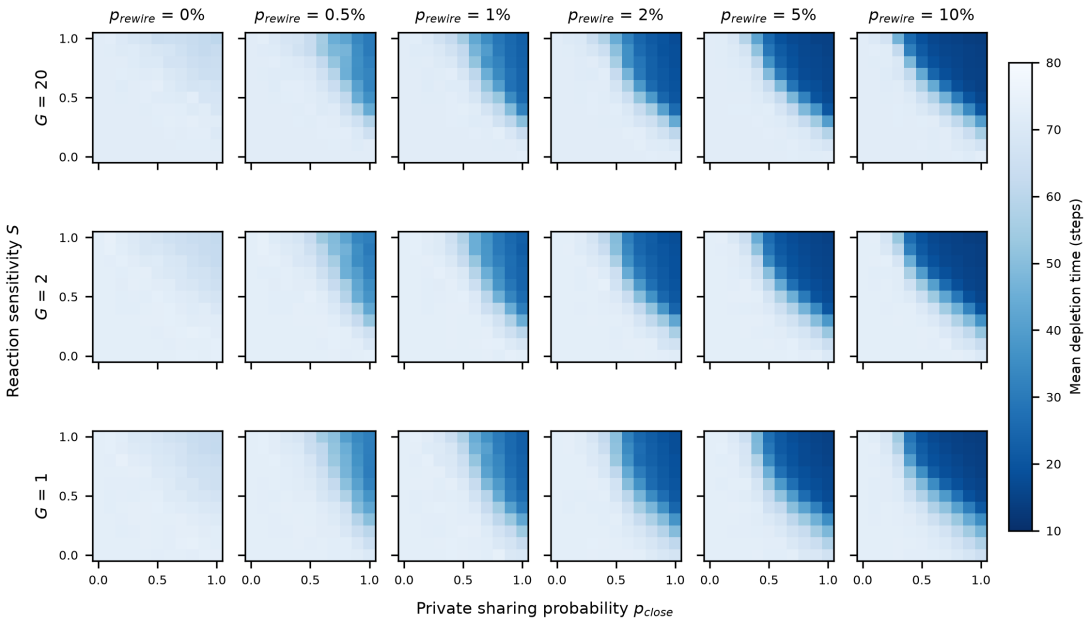


Figure 8: Mean depletion time (steps) at steepness $G = 20$, 2, and 1, at selected rewiring probabilities. Each cell shows the mean of 8 ($G = 20$) or 10 ($G = 1, 2$) matched runs.